

Bài tập lần 14

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5.15

$$(I) \begin{cases} U_{tt} - c^2 U_{xx} = F(x, t), & 0 < x < L, t > 0 \\ U_x(0, t) = t^2, & U(L, t) = -t, & t \geq 0 \\ U(x, 0) = x^2 - L^2 \\ U_t(x, 0) = \sin^2 \frac{\pi x}{L}, & 0 \leq x \leq L \end{cases}$$

- Giả sử u_1 và u_2 là 2 nghiệm của (I).

Đặt $v = u_1 - u_2$. Ta sẽ CMR $v \equiv 0$.

Ta có $v_{tt} = u_{1tt} - u_{2tt}$

$$= (c^2 u_{1xx} + F) - (c^2 u_{2xx} + F)$$

$$= c^2 (u_{1xx} - u_{2xx}) = c^2 v_{xx}, \quad 0 < x < L, t > 0$$

$$v_x(x, t) = u_{1x}(x, t) - u_{2x}(x, t)$$

$$\Rightarrow v_x(0, t) = u_{1x}(0, t) - u_{2x}(0, t) = t^2 - t^2 = 0$$

$$v(L, t) = u_1(L, t) - u_2(L, t) = -t + t = 0$$

$$v(x, 0) = u_1(x, 0) - u_2(x, 0) = (x^2 - L^2) - (x^2 - L^2) = 0$$

$$v_t(x, 0) = u_{1t}(x, 0) - u_{2t}(x, 0) = \sin^2 \frac{\pi x}{L} - \sin^2 \frac{\pi x}{L} = 0$$

$$\Rightarrow \begin{cases} v_{tt} = c^2 v_{xx} & 0 < x < L, t > 0 \\ v_x(0, t) = v(L, t) = 0 & t \geq 0 \\ v(x, 0) = v_t(x, 0) = 0 & 0 \leq x \leq L \end{cases}$$

$$\text{Xét } I(t) = \frac{1}{2} \int_0^L [v_t^2(x, t) + c^2 v_x^2(x, t)] dx.$$

$$I(0) = \frac{1}{2} \int_0^L [v_t^2(x, 0) + c^2 v_x^2(x, 0)] dx = \frac{1}{2} \int_0^L [0^2 + c^2 0^2] dx = 0$$

$$I'(t) = \int_0^L (v_t v_{tt} + c^2 v_x v_{xt}) dx$$

$$= \int_0^L (c^2 v_t v_{xx} - c^2 v_x v_{xt}) dx = c^2 \int_0^L (v_t v_x)' dx$$

$$= c^2 v_t v_x \Big|_{x=0}^{x=L}$$

$$= c^2 [v_t(L,t) v_x(L,t) - v_t(0,t) v_x(0,t)] = 0$$

$$(D_0 \quad v(L,t) = v_x(0,t) = 0 \Rightarrow v_t(L,t) = v_{xt}(0,t) = 0)$$

$$\rightarrow I(t) = I(0) = \int_0^L v_t^2 + c^2 v_x^2 dx$$

$$= \frac{1}{2} \int_0^L (v_t^2 + c^2 v_x^2) dx = 0$$

$$\rightarrow v_t^2 + c^2 v_x^2 = 0 \quad \forall x \in [0, L]$$

$$\rightarrow v_t = v_x = 0 \rightarrow v = \text{const} = v(x, 0) = 0$$

$$\rightarrow u_1 = u_2$$

$\rightarrow$ Bài toán (I) có nghiệm duy nhất u (nếu u là nghiệm của bài toán)

5.16

$$(I) \begin{cases} u_{tt} + u_t - c^2 u_{xx} = 0 & a < x < b, t > 0 \\ u(a, t) = u_x(b, t) = 0 & t \geq 0 \\ u(x, 0) = f(x) & a \leq x \leq b \\ u_t(x, 0) = g(x) & a \leq x \leq b \end{cases}$$

Giả sử u_1 và u_2 là 2 nghiệm của (I)

Đặt $v = u_1 - u_2$. Ta sẽ CMR $v \equiv 0$

$$\text{Ta có } v_{tt} = u_{1tt} - u_{2tt}$$

$$= (c^2 u_{1xx} - u_{1t}) - (c^2 u_{2xx} - u_{2t})$$

$$= c^2 (u_{1xx} - u_{2xx}) - (u_{1t} - u_{2t})$$

$$= c^2 v_{xx} - v_t$$

$$\rightarrow v_{tt} + v_t = c^2 v_{xx} \quad (a < x < b, t > 0)$$

$$v(a, t) = u_1(a, t) - u_2(a, t) = 0$$

$$v_x(b, t) = u_{1x}(b, t) - u_{2x}(b, t) = 0$$

$$v(x, 0) = u_1(x, 0) - u_2(x, 0) = f(x) - f(x) = 0$$

$$u(x,0) = u_1(x,0) - u_2(x,0) = g(x) - g(x) = 0$$

$$\begin{cases} v_{tt} + v_t = c^2 v_{xx} & a < x < b, t > 0 \\ v(a,t) = v_x(b,t) = 0 & t \geq 0 \end{cases}$$

$$v(x,0) = v_t(x,0) = 0 \quad a \leq x \leq b$$

$$\text{Xét } I(t) = \frac{1}{2} \int_a^b (v_t^2 + c^2 v_x^2) dx$$

$$I(0) = \frac{1}{2} \int_a^b (v_t^2(x,0) + c^2 v_x^2(x,0)) dx = 0$$

$$(D_0 \quad v(x,0) = v_t(x,0) = 0 \rightarrow v_x(x,0) = v_t(x,0) = 0)$$

$$I'(t) = \int_a^b (v_t v_{tt} + c^2 v_x v_{xt}) dx$$

$$= \int_a^b v_t (c^2 v_{xx} - v_t + c^2 v_x v_{xt}) dx = \int_a^b [c^2 (v_t v_{xx} + v_x v_{xt}) - v_t^2] dx$$

$$= c^2 \int_a^b (v_t v_{xx} + v_x v_{xt}) dx - \int_a^b (v_t)^2 dx$$

$$= c^2 (v_t v_x) \Big|_{x=a}^{x=b} - \int_a^b (v_t)^2 dx$$

$$= c^2 v_t(b,t) v_x(b,t) - c^2 v_t(a,t) v_x(a,t) - \int_a^b (v_t)^2 dx = 0 - 0 - \int_a^b (v_t)^2 dx$$

$$(D_0 \quad v(a,t) = v_x(b,t) = 0 \rightarrow v_t(a,t) = v_x(b,t) = 0)$$

$$= - \int_a^b (v_t)^2 dx \leq 0$$

$$\rightarrow 0 \leq I(t) \leq I(0) = 0 \rightarrow I(t) = 0 = \frac{1}{2} \int_a^b (v_t^2 + c^2 v_x^2) dx \quad \forall x \in [a,b]$$

$$\rightarrow v_t = v_x = 0 \rightarrow v = \text{const} = v(x,0) = 0$$

$$\rightarrow u_1 = u_2$$

$\rightarrow$ Nếu u là nghiệm của (I) thì u là duy nhất.

5.17

$$\begin{cases}
 U_{tt} - c^2 U_{xx} + hU = F(x,t) & -\infty < x < +\infty, t > 0, h > 0 \\
 \lim_{x \rightarrow \pm\infty} U(x,t) = \lim_{x \rightarrow \pm\infty} U_x(x,t) = \lim_{x \rightarrow \pm\infty} U_t(x,t) = 0 & t \geq 0 \\
 \int_{-\infty}^{+\infty} (U_t^2 + c^2 U_x^2 + hU^2) dx < \infty, t \geq 0 \\
 U(x,0) = f(x) & -\infty < x < +\infty \\
 U_t(x,0) = g(x)
 \end{cases}$$

- Giả sử U_1 và U_2 là nghiệm của (I), Ta có
Đặt $V = U_1 - U_2$ Ta có $V \equiv 0$

$$\begin{aligned}
 \text{Ta có } V_{tt} &= U_{1tt} - U_{2tt} \\
 &= c^2 U_{1xx} - hU_1 + F - c^2 U_{2xx} + hU_2 - F \\
 &= c^2 (U_{1xx} - U_{2xx}) - (hU_1 - hU_2) \\
 &= c^2 V_{xx} - hV
 \end{aligned}$$

$$V_{tt} + hV - c^2 V_{xx} = 0$$

$$\lim_{x \rightarrow \pm\infty} V(x,t) = \lim_{x \rightarrow \pm\infty} (U_1(x,t) - U_2(x,t)) = 0$$

$$\lim_{x \rightarrow \pm\infty} V_x(x,t) = \lim_{x \rightarrow \pm\infty} (U_{1x}(x,t) - U_{2x}(x,t)) = 0$$

$$\lim_{x \rightarrow \pm\infty} V_t(x,t) = \lim_{x \rightarrow \pm\infty} (U_{1t}(x,t) - U_{2t}(x,t)) = 0$$

$$V(x,0) = U_1(x,0) - U_2(x,0) = f(x) - f(x) = 0$$

$$V_t(x,0) = U_{1t}(x,0) - U_{2t}(x,0) = g(x) - g(x) = 0$$

$$\begin{cases}
 V_{tt} - c^2 V_{xx} + hV = 0 & -\infty < x < +\infty, t > 0 \\
 \lim_{x \rightarrow \pm\infty} V(x,t) = \lim_{x \rightarrow \pm\infty} V_x(x,t) = \lim_{x \rightarrow \pm\infty} V_t(x,t) = 0 & t \geq 0 \\
 V(x,0) = V_t(x,0) = 0 & -\infty < x < +\infty
 \end{cases}$$

$$\text{Xét } I(t) = \frac{1}{2} \int_{-\infty}^{+\infty} (v_t^2 + c^2 v_x^2 + \hbar v^2) dx$$

$$I(0) = \frac{1}{2} \int_{-\infty}^{+\infty} (v_t^2(x,0) + c^2 v_x^2(x,0) + \hbar v^2(x,0)) dx = 0$$

$$I'(t) = \int_{-\infty}^{+\infty} (v_t v_{tt} + c^2 v_x v_{xt} + \hbar v v_t) dx$$

$$= \int_{-\infty}^{+\infty} (v_t (-\hbar v + c^2 v_{xx}) + c^2 v_x v_{xt} + \hbar v v_t) dx$$

$$= c^2 \int_{-\infty}^{+\infty} (v_t v_{xx} + v_x v_{xt}) dx = c^2 \int_{-\infty}^{+\infty} (v_t v_x)' dx$$

$$= \lim_{\xi \rightarrow +\infty} \lim_{\mu \rightarrow -\infty} c^2 (v_t v_x) \Big|_{x=\mu}^{x=\xi}$$

$$= \lim_{\xi \rightarrow +\infty} \lim_{\mu \rightarrow -\infty} c^2 [v_t(\xi, t) v_x(\xi, t) - v_t(\mu, t) v_x(\mu, t)]$$

$$= 0 \quad \text{Do } \lim_{x \rightarrow \pm \infty} v_x(x, t) = 0$$

$$\Rightarrow I'(t) = (\text{const}) = I(0) = 0$$

$$\Rightarrow I(t) = (\text{const}) = I(0) = 0$$

$$\Rightarrow \frac{1}{2} \int_{-\infty}^{+\infty} (v_t^2 + c^2 v_x^2 + \hbar v^2) dx = 0$$

$$\Rightarrow v_t^2 + c^2 v_x^2 + \hbar v^2 = 0$$

$$\Rightarrow v_t = v_x = v = 0$$

$$\Rightarrow u_1 = u_2$$

$\Rightarrow$ Kết luận: Nếu u là 1 n.c.m (1) thì u là duy nhất

5.18

$$(I) \begin{cases} u_t - k u_{xx} = F(x,t), & 0 < x < L, t > 0 \\ u(0,t) - \alpha u_x(0,t) = a(t) & t \geq 0 \\ u(L,t) + \beta u_x(L,t) = b(t) \\ u(x,0) = f(x) & 0 < x < L \end{cases}$$

- Giả sử u_1 và u_2 là 2 h° của (I)

thì $v = u_1 - u_2$. Ta có $v \equiv 0$

Ta có $v_t = u_{1t} - u_{2t}$

$$= k u_{1xx} + F(x,t) - k u_{2xx} - F(x,t)$$

$$= k (u_{1xx} - u_{2xx}) = k v_{xx}$$

$$v(0,t) = u_1(0,t) - u_2(0,t)$$

$$= \alpha u_{1x}(0,t) + a(t) - \alpha u_{2x}(0,t) - a(t)$$

$$= \alpha (u_{1x}(0,t) - u_{2x}(0,t)) = \alpha v_x(0,t)$$

$$v(L,t) = u_1(L,t) - u_2(L,t)$$

$$= b(t) - \beta u_{1x}(L,t) - b(t) + \beta u_{2x}(L,t)$$

$$= -\beta v_x(L,t)$$

$$v(x,0) = u_1(x,0) - u_2(x,0) = f - f = 0$$

$$\Rightarrow \begin{cases} v_t = k v_{xx} & 0 < x < L, t > 0 \\ v(0,t) = \alpha v_x(0,t), v(L,t) = -\beta v_x(L,t) & t \geq 0 \\ v(x,0) = 0 & 0 < x < L \end{cases}$$

$$\text{Xét } I(t) = \frac{1}{2} \int_0^L v^2(x,t) dx$$

$$I(0) = \frac{1}{2} \int_0^L v^2(x,0) dx = 0$$

$$I'(t) = \int_0^L v v_t dx = k \int_0^L v v_{xx} dx$$

$$= k \int_0^L v dv = k \left(v^2 \Big|_{x=0}^{x=L} - \int_0^L (v_x)^2 dx \right)$$

$$= k \left(v(L,t)v_x(L,t) - v(0,t)v_x(0,t) - \int_0^L v_x^2(x,t) dx \right)$$

$$= k \left(-\beta v_x^2(L,t) - \alpha v_x^2(0,t) - \int_0^L v_x^2(x,t) dx \right) \quad (\alpha, \beta \geq 0, k > 0)$$

$$\leq 0$$

$$\Rightarrow 0 \leq I(t) \leq I(0) = 0$$

$$\Rightarrow I(t) = 0 = \frac{1}{2} \int_0^L v^2(x,t) dx$$

$$\Rightarrow v(x,t) = 0 \Rightarrow u_1 = u_2$$

$\Rightarrow$ Nếu u là nghiệm của (I) thì u là duy nhất.

(Câu 3 ý (a) đề 3 - GK - K64 TT)

$$(I) \begin{cases} u_t(x,t) = 3u_{xx}(x,t) - 6u_x(x,t) + e^x \sin x & 0 < x < \pi, t > 0 \\ u(0,t) = u(\pi,t) = 0 & t \geq 0 \\ u(x,0) = xe^x & 0 \leq x \leq \pi \end{cases}$$

- Giả sử u_1 và u_2 là 2 nghiệm của (I)

Đặt $v = u_1 - u_2$ Ta có $v \equiv 0$

Ta có $v_t = u_{1t} - u_{2t} =$

$$\begin{aligned} &= 3u_{1xx} - 6u_{1x} + e^x \sin x - 3u_{2xx} + 6u_{2x} - e^x \sin x \\ &= 3(u_{1xx} - u_{2xx}) - 6(u_{1x} - u_{2x}) \\ &= 3v_{xx} - 6v_x \end{aligned}$$

$$\Rightarrow v_t = 3v_{xx} - 6v_x$$

$$v(0,t) = u_1(0,t) - u_2(0,t) = 0$$

$$v(\pi,t) = u_1(\pi,t) - u_2(\pi,t) = 0$$

$$v(x,0) = u_1(x,0) - u_2(x,0) = xe^x - xe^x = 0$$

$$\begin{aligned} \int v_t &= 3v_{xx} - 6v_x \quad 0 < x < \pi, t > 0 \\ \rightarrow \begin{cases} v(0, t) = v(\pi, t) = 0 & t \geq 0 \\ v(x, 0) = 0 & 0 \leq x \leq \pi \end{cases} \end{aligned}$$

$$\text{Xét } I(t) = \frac{1}{2} \int_0^\pi v^2(x, t) dx$$

$$\rightarrow I(0) = \frac{1}{2} \int_0^\pi v^2(x, 0) dx = 0$$

$$\begin{aligned} I'(t) &= \int_0^\pi v v_t dx = \int_0^\pi v (3v_{xx} - 6v_x) dx \\ &= 3 \int_0^\pi v v_{xx} dx - 6 \int_0^\pi v v_x dx \end{aligned}$$

$$\begin{aligned} \text{Xét } I_1(t) &= 3 \int_0^\pi v(x, t) v_{xx}(x, t) dx \\ &= 3 \left(v(x, t) v_x(x, t) \Big|_0^\pi - \int_0^\pi v_x^2(x, t) dx \right) \\ &= 3 \left(v(\pi, t) v_x(\pi, t) - v(0, t) v_x(0, t) - \int_0^\pi v_x^2(x, t) dx \right) \\ &= -3 \int_0^\pi v_x^2(x, t) dx \end{aligned}$$

$$\begin{aligned} \text{Xét } I_2(t) &= 6 \int_0^\pi v(x, t) v_x(x, t) dx \\ &= 6 \left(v^2(x, t) \Big|_0^\pi - \int_0^\pi v(x, t) v_{xx}(x, t) dx \right) \\ &= 6 (v^2(\pi, t) - v^2(0, t)) - 6 I_1(t) \end{aligned}$$

$$\rightarrow 7 I_2(t) = 0 \rightarrow I_2(t) = 0$$

$$\begin{aligned} \rightarrow I'(t) &= I_1(t) - I_2(t) \\ &= -3 \int_0^{\pi} v_x^2(x, t) dx \leq 0 \end{aligned}$$

$$\rightarrow 0 \leq I(t) \leq I(0) = 0$$

$$\Rightarrow I(t) = 0 = \frac{1}{2} \int_0^{\pi} v^2(x, t) dx$$

$$\rightarrow v(x, t) = 0 \rightarrow u_1 = u_2$$

$\rightarrow$ Vậy bài toán (I) có tối đa 1 nghiệm

Câu 3 - y' (a) đề 5 - GK - K64Π

$$(I) \begin{cases} U_{tt} = U_{xx} - U + F(x, t) & 0 < x < 2, t > 0 \\ U_x(0, t) = 2, \quad U(2, t) = e^2 + 1 & t \geq 0 \\ U(x, 0) = e^x + 1, \quad U_t(x, 0) = \cos^3 \frac{\pi x}{4} & 0 \leq x \leq 2 \end{cases}$$

- Giả sử u_1 và u_2 là 2 n° của (I)

Đặt $v = u_1 - u_2$. Ta có CMR $v \equiv 0$

$$\text{Ta có } v_{tt} = u_{1tt} - u_{2tt}$$

$$= u_{1xx} - u_1 + F(x, t) - u_{2xx} + u_2 - F(x, t)$$

$$= (u_{1xx} - u_{2xx}) - (u_1 - u_2) = v_{xx} - v$$

$$v_x(0, t) = u_{1x}(0, t) - u_{2x}(0, t) = 2 - 2 = 0$$

$$v(2, t) = u_1(2, t) - u_2(2, t) = e^2 + 1 - e^2 - 1 = 0$$

$$v(x, 0) = u_1(x, 0) - u_2(x, 0) = e^x + 1 - e^x - 1 = 0$$

$$v_t(x, 0) = u_{1t}(x, 0) - u_{2t}(x, 0) = \cos^3 \frac{\pi x}{4} - \cos^3 \frac{\pi x}{4} = 0$$

$$\rightarrow \begin{cases} v_{tt} = v_{xx} - v & 0 < x < 2, t > 0 \\ v_x(0, t) = v(2, t) = 0 & t \geq 0 \\ v(x, 0) = v_t(x, 0) = 0 & 0 \leq x \leq 2 \end{cases}$$

$$\text{Xét } I(t) = \frac{1}{2} \int_0^2 (v_t^2 + v_x^2 + v^2) dx$$

$$\rightarrow I(t) = \frac{1}{2} \int_0^2 (u_t^2(x,t) + u_x^2(x,t) + v^2(x,t)) dx = 0$$

$$I'(t) = \int_0^2 (u_t u_{tt} + u_x v_{xt} + u v_t) dx$$

$$= \int_0^2 (u_t (v_{xx} - u) + u_x v_{xt} + u v_t) dx$$

$$= \int_0^2 (u_t v_{xx} + u_x v_{xt}) dx = \int_0^2 (v_t v_x)' dx$$

$$= v_t v_x \Big|_{x=0}^{x=2} = v_t(2,t) v_x(2,t) - v_t(0,t) v_x(0,t) = 0$$

$$Do \quad v_x(0,t) = u(2,t) = 0 \Rightarrow v_x(0,t) = v_t(2,t) = 0$$

$$\Rightarrow I(t) = I(0) = 0 = \frac{1}{2} \int_0^2 (u_t^2 + u_x^2 + v^2) dx$$

$$\rightarrow u_t = u_x = v = 0$$

$$\Rightarrow M_1 = M_2$$

$\Rightarrow$ Bảo toàn (I) có tại t là 1 n°